

ABC by Example and Experiments (Top-Down Approach)

Pointers!

```
@ <stdio.h>
```

```
fn main()
```

```
{
```

```
    local a: int;
```

```
    a = 1;
```

```
    printf("a = %d\n", a);
```

```
}
```

```
@ <stdio.h>
```

```
fn main()  
{  
    local a: int;  
  
    a = 1;  
  
    printf("a = %d\n", a);  
    printf("&a = %p\n", &a);  
    printf("sizeof(a) = %zu\n", sizeof(a));  
  
}
```

@ <stdio.h>

```
fn main()
{
    local a: int;
    local ptr: -> int;

    a = 1;

    printf("a = %d\n", a);

    ptr = &a;

    *ptr = 42;
    printf("a = %d\n", a);
}
```

```
@ <stdio.h>
```

```
fn foo(n: int)
{
    n = 42;
}
```

```
fn main()
{
    local a: int = 123;

    printf("a = %d\n", a);
    foo(a);
    printf("a = %d\n", a);
}
```

@ <stdio.h>

```
fn foo(n: -> int)
{
    *n = 42;
}
```

```
fn main()
{
    local a: int = 123;

    printf("a = %d\n", a);
    foo(&a);
    printf("a = %d\n", a);
}
```

```
@ <stdio.h>
```

```
fn main()  
{  
    local a: int;  
    local ptr: -> int;  
  
    a = 1;  
  
    printf("a = %d\n", a);  
  
    ptr = &a;  
  
    *ptr = 42;  
    printf("a = %d\n", a);  
}
```

@ <stdio.h>

```
fn main()
{
    local a: int;
    local ptr: -> int;

    a = 1;

    printf("a = %d\n", a);
    printf("&a = %p\n", &a);
    printf("sizeof(a) = %zu\n", sizeof(a));

    ptr = &a;
    printf("ptr = %p\n", ptr);
    printf("sizeof(ptr) = %zu\n", sizeof(ptr));

    *ptr = 42;
    printf("a = %d\n", a);
}
```